

Accessibility Structure in the Rubik's Cube Representation

Lie-Generated Accessibility, Hybrid Sectors, and T7 Morphisms

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This paper is Part III of the RIME trilogy. Paper I established the spectral decomposition and primitive sector structure. Paper II characterized the transport topology and M_2 Principle. The present paper develops Lie accessibility and proves the T7 Theorem.

Abstract

Problem. When a finite group G acts on a representation $V = V_A \oplus V_B$, what is the structural difference between *infinitesimal generation* (the Lie algebra $\mathcal{L} = \text{Lie}\{A_g\}$ with $A_g = \log \rho(g)$) and *discrete composition* of the original group action? Are there morphisms reachable by composition that no commutator depth of $\mathcal{L}$ can produce?

Approach. Every element of the universal enveloping algebra $\mathcal{U}(\mathcal{L})$ is block-diagonal (Lemma 4.1, Lie-Generated Support Invariance), so $P_\alpha X P_\beta = 0$ for all $X \in \mathcal{U}(\mathcal{L})$ whenever α, β are supported in disjoint blocks. Lie-generated accessibility is block-preserving — not just at finite commutator depth, but at all algebraic depths. Discrete composition $P_\alpha \rho(g) P_\gamma \rho(h) P_\beta$ through a *hybrid sector* γ spanning both blocks is not an element of $\mathcal{U}(\mathcal{L})$ — it is a projector-mediated morphism outside Lie-generated accessibility.

Results. Under conditions C0–C3 — center incompleteness ($Z \subsetneq \text{Comm}(\rho)$), shared non-commutative support, transport-active hybrid sectors, and block-preserving Lie dynamics — T7 morphisms arise: cross-block morphisms reachable by discrete composition through hybrid sectors but inaccessible to Lie-generated accessibility at every commutator depth. C2 and C3 are proved necessary; C1 necessity is conjectured for general non-abelian groups and supported by all small-group systems tested. The 228-dimensional Rubik cube exhibits 5 T7 morphisms, all cross-block; S_3 negative controls (0 T7) isolate C0 as the foundational structural divide.

Implications. In the Rubik cube, compositional accessibility is strictly richer than Lie-generated accessibility: the continuous limit is structurally incomplete, omitting an entire class of cross-block morphisms that the discrete group action realizes. The S_3 negative controls support this: their canonical decompositions fail C0 (center completeness forces diagonal K), consistent with C0–C3 as non-trivial characterizing conditions. Formally, compositional accessibility strictly contains Lie-generated accessibility, and the T7 morphisms are the cross-block morphisms outside Lie-generated accessibility (§7).

Core claim. The trilogy establishes a single structural fact across three papers: **projector geometry $\not\subseteq$ Lie tangent geometry**. [2] established that projector geometry exists (6 spectral layers, 9 primitive sectors). [1] characterized the noncommutative curvature within it (M_2 principle, transport topology). This paper proves that compositional accessibility strictly exceeds Lie-generated accessibility (T7 morphisms). The spectral projector geometry of the representation is not contained in the Lie tangent geometry generated by $\{A_g = \log \rho(g)\}$ — a structural consequence of Schur's lemma and the isotypic decomposition (Layers A+B), verified on the Rubik's cube (228-dim); two S_3 systems (9-dim and 12-dim) serve as negative controls exhibiting no T7 morphisms.

Notation Table

Symbol	Meaning	Origin
$A, P_\alpha, K_{\alpha\beta}, \text{Supp}_{\text{nc}},$ blocks, layers, sectors S1–S9	Spectral objects, transport graph, noncommutative support	Papers I–II
$A_g = \log \rho(g)$	Infinitesimal generator — logarithmic embedding of discrete generator	this paper
$\mathcal{A} = \text{span}\{\rho(g)\}$	Image of the group algebra on V — block-diagonal by the isotypic decomposition	this paper
$\mathcal{L} = \text{Lie}\{A_g\}$	Lie algebra generated by $\{A_g\}_{g \in S}$; contained in $\mathcal{A}$ (Layer A, 2.3)	this paper
$\mathcal{U}(\mathcal{L})$	Universal enveloping algebra of $\mathcal{L}$ — the full associative algebra of Lie polynomials	this paper
$\kappa_0(\alpha, \beta) = \max_g \ P_\alpha A_g P_\beta\ _F$	Lie gradient (κ_0) — single-generator morphism within Lie-generated accessibility	this paper
$\kappa_1(\alpha, \beta) = \max_{g,h} \ P_\alpha [A_g, A_h] P_\beta\ _F$	Lie curvature (κ_1) — commutator-mediated morphism within Lie-generated accessibility	this paper
$\kappa_d(\alpha, \beta)$	Lie-generated accessibility at commutator depth d	this paper
T7 morphism	Cross-block morphism outside Lie-generated accessibility: $K_{\alpha\beta} = 0, \kappa_d(\alpha, \beta) = 0$ for all d , yet $\alpha \rightsquigarrow \beta$ via composition through a hybrid sector	this paper
Hybrid sector	A primitive sector whose projector spans multiple blocks; when transport-active, it enables T7 morphisms across block boundaries (C2)	Paper II/III
C0	Center Incompleteness: $Z \subsetneq \text{Comm}(\rho)$ — the transport-generated center is a proper subalgebra of the full commutant; sectors aggregate isotypic components, enabling off-diagonal transport	Paper II
C1, C2, C3	T7 characterizing conditions: (C1) shared isotypic support, (C2) transport-active hybrid sector, (C3) block-preserving dynamics	this paper
$\mathcal{T}, \bar{\mathcal{T}}$	Transport category and its compositional accessibility closure	this paper

Full notation glossary: `docs/conventions.md` and (CCS Part 0).

1 Introduction

Conventions. We use a right-handed Cartesian coordinate system ($+X \rightarrow R, +Y \rightarrow U, +Z \rightarrow F$). The representation space $V = \mathbb{C}^{228}$ decomposes as $V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}}$ ($64 + 144 + 8 + 12$). The generator set S is the 18 standard face-turn generators ($S = S^{-1}$). Full conventions: `docs/conventions.md`. All numerical values: (CCS-I §2, CCS-III §10.4, Appendices F–I).

1.1 Where Papers I and II Left Off

[2] established the spectral ontology: the averaging operator $A = \frac{1}{|S|} \sum \rho(s)$ has a rational 6-layer spectrum, refines into 9 primitive sectors under $\text{Center}\{A, \text{QT}_{\text{all}}, \text{HT}_{\text{all}}\}$, and is organized by a refinement semilattice. [1] established the static transport topology: $K_{\alpha\beta} = \max_g \|P_\alpha \rho(g) P_\beta\|_F > 0$ when two sectors share noncommutative support (sufficient condition; one Type II exception, $S8 \leftrightarrow S9$), with S6 as the primary hub and S1 as isolated. Both papers study static objects — the spectral decomposition and the transport graph.

Paper III studies **dynamics**. Given the transport topology, what can actually reach what — and at what Lie-algebraic depth?

1.2 The Central Question

Take two primitive sectors α and β supported in disjoint blocks of the representation. Can a nonzero morphism exist from β to α ?

Mechanism	Object	Cross-block?
Direct	$K_{\alpha\beta} = \max_g \ P_\alpha \rho(g) P_\beta\ _F$	No — Type I via shared Supp_nc; one Type II exception (S8↔S9) via CP channel ([1])
Lie gradient (κ_0)	$\kappa_0(\alpha, \beta) = \max_g \ P_\alpha A_g P_\beta\ _F$	No — $A_g = \log \rho(g)$ is block-diagonal
Lie curvature (κ_1)	$\kappa_1(\alpha, \beta) = \max \ P_\alpha [A_g, A_h] P_\beta\ _F$	No — $[A_g, A_h]$ is block-diagonal
Composition (T7)	$P_\alpha \rho(g) P_\gamma \rho(h) P_\beta$	Yes — through a hybrid sector γ

The first three rows are structural consequences of Schur’s lemma (Lemmas 0–1, Layers A+B). $A_g = \log \rho(g)$ inherits the block-diagonal structure of $\rho(g)$. Commutators of block-diagonal matrices are block-diagonal. By induction, **every element of $\mathcal{U}(\mathcal{L})$ — the full universal enveloping algebra — is block-diagonal**. Lie-generated accessibility is structurally blind to cross-block morphisms.

The fourth row — discrete composition through a hybrid sector — is not blocked. A hybrid sector γ whose projector spans both blocks can route a composition path: $\alpha \rightarrow \gamma \rightarrow \beta$. The discrete group action preserves this path; the Lie algebra collapses it.

This is the discrete/continuous split. $\mathcal{U}(\mathcal{L})$ probes infinitesimal tangent geometry inside blocks; discrete composition probes global projector geometry across blocks. The mismatch between Lie-generated accessibility and compositional accessibility is the conceptual core of this paper. This paper is not a solver: no Kociemba algorithm, no sticker rendering, no neural networks, no genetic algorithms, no control policies. The κ_d hierarchy is a structural diagnostic — it measures reachability, not optimality.

1.3 T7 Morphisms

A **T7 morphism** is a cross-block morphism outside Lie-generated accessibility: $K_{\alpha\beta} = 0$, $\kappa_d(\alpha, \beta) = 0$ for all commutator depths d , yet there exists a finite composition path

$$P_\alpha \rho(g_1) P_{\gamma_1} \cdots \rho(g_n) P_\beta \neq 0$$

through a hybrid sector.

Lie-generated accessibility is block-preserving at every algebraic depth (Layers A+B, Lemmas 0–1). Only compositional accessibility bridges blocks — through projector-mediated morphisms outside Lie-generated accessibility. Four distinctions keep the trilogy’s concepts distinct: (i) **T7** $\neq M_2$ — the Rubik cube exhibits both as structurally orthogonal (§5.5); (ii) **curvature** $\neq$ **transport** — curvature is within-block κ_1 , transport is depth-1 K ([1]); (iii) **transport** $\neq$ **accessibility** — transport is single-generator edges, accessibility is compositional closure; (iv) **center completeness** $\neq$ **sector granularity** — $Z = \text{Comm}(\rho)$ forces diagonal K ; $Z \subsetneq \text{Comm}(\rho)$ enables off-diagonal transport (C0, §5.2).

1.4 T7 Theorem (Preview)

T7 morphisms require four structural conditions: (C0) center incompleteness ($Z \subsetneq \text{Comm}(\rho)$), (C1) shared noncommutative support, (C2) transport-active hybrid sectors, (C3) block-preserving Lie dynamics. The formal theorem is stated and proved below (see the T7 Theorem).

1.5 The S_3 Narrative Arc

S_3 (order 6) serves as the negative control. Under the canonical Center $\{A_{\text{full}}, A_{\text{trans}}\}$ decomposition, neither $\text{nat} \oplus \text{reg}$ (9-dim) nor $\text{reg} \oplus \text{reg}$ (12-dim) exhibits T7 morphisms — both fail C0 ($Z = \text{Comm}(\rho)$ forces diagonal K) and lack transport-active hybrid sectors (C2). These failures confirm C0–C3 as non-trivial characterizing conditions. The Rubik cube (228-dim) satisfies all four conditions and exhibits 5 verified T7 morphisms. Section 6 develops the full comparison.

1.6 Structure

Part	Content	Question
I (Setup and Accessibility Hierarchy)	Lie-generated accessibility: K, κ_0, κ_1	What can reach what, at what Lie-algebraic depth?
II (Why Lie-Generated Accessibility Cannot Cross Blocks)	Lie-Generated Support Invariance (4.1) + formal definitions	Why can't $\mathcal{U}(\mathcal{L})$ cross blocks?
III (The T7 Morphism Theorem)	T7 Morphism Theorem	When does compositional accessibility transcend Lie-generated accessibility?
IV (S_3 Prototypes and Rubik Realization)	S_3 negative controls $\rightarrow$ Rubik at scale	How does this manifest across systems?
V (The Structure Behind T7)	Structural separation	What structure does T7 reveal?
VI (Discussion and Concluding Perspective)	Discussion, concluding perspective	

Figure 1 shows the full trilogy cascade from spectral decomposition through transport topology to Lie accessibility.

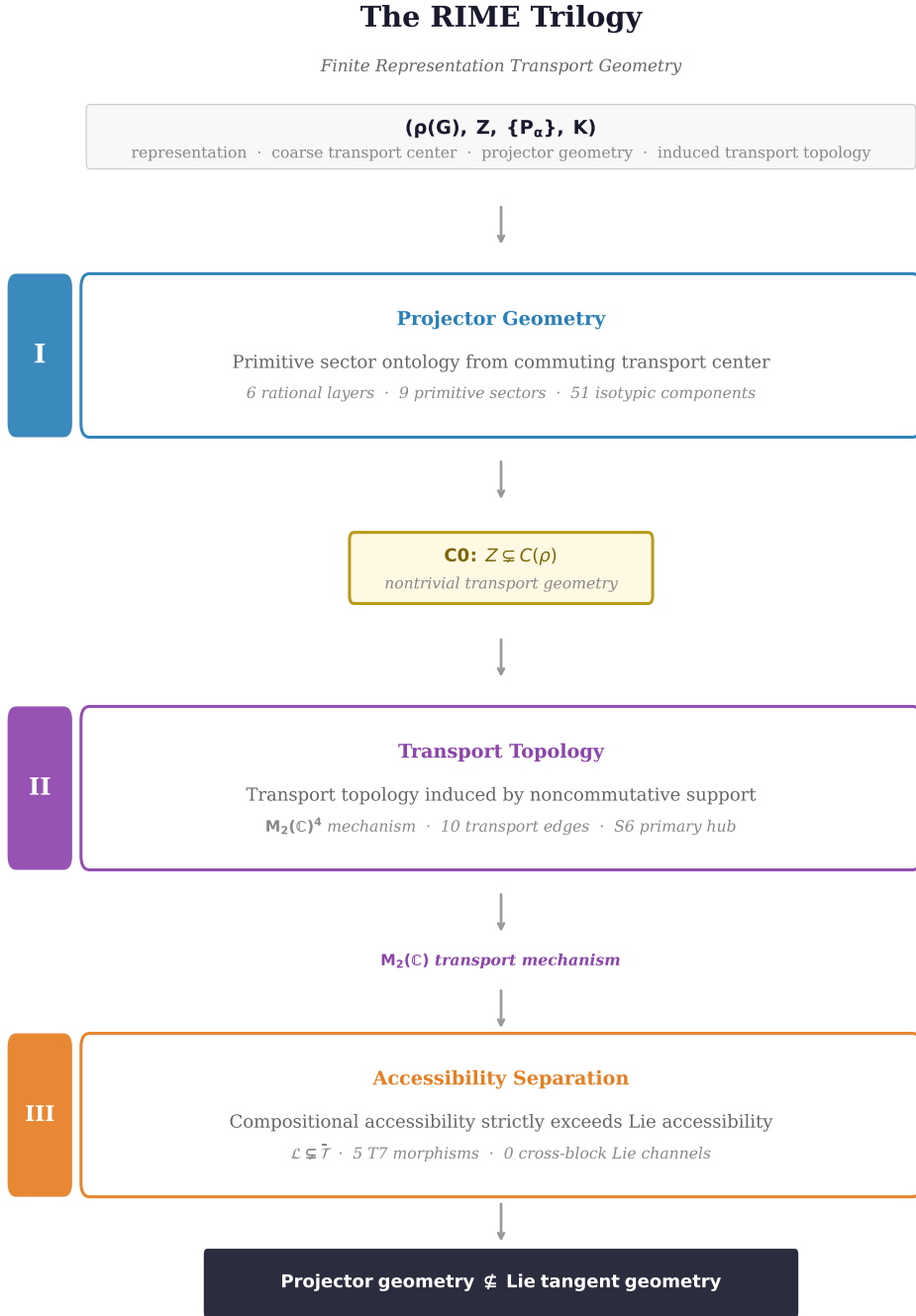


Figure 1. The RIME trilogy cascade: spectral decomposition (Paper I) $\rightarrow$ transport topology (Paper II) $\rightarrow$ Lie accessibility (Paper III), from $\rho(g)$ through A , $\{V_\lambda\}$, $\{S_\alpha\}$, $K_{\alpha\beta}$, κ_0/κ_1 , to T7 morphisms. Each stage refines the structural resolution: layers aggregate isotypic components, sectors aggregate layers, transport connects sectors, and accessibility distinguishes Lie-reachable from composition-only paths. The single structural fact unifying the trilogy is that spectral projector geometry is not contained in Lie tangent geometry.

2 Setup and Prerequisites

2.1 Representation and Generators

Geometric conventions. We use a right-handed Cartesian coordinate system ($+X \rightarrow R$, $+Y \rightarrow U$, $+Z \rightarrow F$). Corner and edge positions are defined by sign vectors in $\{\pm 1\}^3$ and $\{\pm 1, 0\}^3$ respectively; all combinatorial constants (orderings, face adjacency, rotation strips) are derived algorithmically from these coordinate rules, not hand-enumerated. Orderings are induced canonically by clockwise XZ-plane cycles. Clockwise rotation is defined relative to the outward face normal. Full detail: `docs/conventions.md`.

We work with the 228-dimensional representation $\rho : G \rightarrow \text{GL}(228, \mathbb{R})$ of the Rubik's cube group, decomposing into four G -invariant blocks:

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}}, \quad 64 + 144 + 8 + 12 = 228,$$

laid out in $\text{cp} \rightarrow \text{ep} \rightarrow \text{co} \rightarrow \text{eo}$ order. By construction, $\rho(g)$ is block-diagonal for all $g \in G$ — each block transforms independently, and cross-block matrix elements are identically zero.

The generator set S consists of the 18 standard face-turn generators ($S = S^{-1}$). The logarithmic embedding $A_g = \log \rho(g)$ inherits the block-diagonal structure (Lemma 4.1, see Lie-Generated Support Invariance below).

All numerical data in this paper correspond to the 6 canonical layers ($k \in \{0, 1, 2, 3, 4, 6\}$, $\lambda = 1 - k/9$) and 9 primitive sectors from the Center $\{A_{18}, \text{QT}_{\text{all}}, \text{HT}_{\text{all}}\}$ joint diagonalization. The single source of truth is the Unified Computational Supplement (CCS).

2.2 Spectral Decomposition

The averaging operator $A = \frac{1}{18} \sum_{s \in S} \rho(s)$ has eigendecomposition:

$$A = \sum_i \lambda_i P_i, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_i P_i = I_{228}$$

with 6 distinct eigenvalues and 9 primitive sectors (Center joint diagonalization). Full tables in (CCS-I §1.3–§1.4).

2.3 Block Decomposition and Supp_nc

Each sector has well-defined block support. The noncommutativity hierarchy ([1]):

$$\|[QT^0, QT^1]\|_F = 2.92 \text{ total}; \quad \text{cp} = 0, \text{ ep} = 2.74 (93.9\%), \text{ co} = 0.61, \text{ eo} = 0.79$$

$$\text{Supp}_{\text{nc}}(\alpha) = \{b \in \{\text{cp}, \text{ep}, \text{co}, \text{eo}\} : P_\alpha|_b \neq 0 \text{ and } \|[QT^0, QT^1]\|_b > 0\}.$$

2.4 Lie Generators

The Lie generators are computed via the principal matrix logarithm:

$$A_g = \log \rho(g)$$

using `scipy.linalg.logm`. Fidelity: $\max_g \|\exp(A_g) - \rho(g)\| < 3 \times 10^{-8}$. The A_g are skew-Hermitian to numerical precision. The full Lie algebra is $\mathcal{L} = \text{Lie}\{A_g\}_{g \in S}$, generated by iterated commutators.

2.5 Hybrid Sectors — A Formal Definition

The term “hybrid sector” has been used informally throughout the trilogy to designate primitive sectors whose projectors span multiple invariant blocks. It is now formalized as a mathematical object.

Definition 2.1 (Hybrid Sector).

Let $V = V_A \oplus V_B \oplus \dots$ be the block decomposition of the representation. A primitive sector E_γ with projector P_γ is called **hybrid** if its projector has nonzero restriction to more than one invariant block:

$$P_\gamma|_{V_A} \neq 0 \quad \text{and} \quad P_\gamma|_{V_B} \neq 0$$

for at least two distinct blocks $A \neq B$.

A hybrid sector is **transport-active** if it participates in nonzero transport morphisms to sectors in both blocks:

$$\exists \alpha \in V_A, \beta \in V_B : K_{\alpha\gamma} > 0 \text{ and } K_{\gamma\beta} > 0$$

Equivalently (computable criterion): $\text{Supp}(E_\gamma) \cap \text{Supp}(E_\alpha) \neq \emptyset$ and $\text{Supp}(E_\gamma) \cap \text{Supp}(E_\beta) \neq \emptyset$, where $\text{Supp}(E) = \{\tau \in \hat{G} : \text{Tr}(P\Pi_\tau) > 0\}$ is the isotypic support.

A hybrid sector that fails the transport-active criterion — hybrid by block support but sharing no irreducible content with sectors in both blocks — is called **inert**. Inert hybrids arise from eigenvalue coincidence in the commutative center Z without shared irrep geometry, and provide no T7 morphism (confirmed by the S_3 inert hybrid counterexample, Appendix C).

Remark 2.2 (Examples — Rubik’s cube, 9-sector resolution).

Sector	Blocks	Transport-active?	Role
S1 (V_1 , 20-dim)	cp+ep	No — isolated	G -invariant subrepresentation
S3 ($V_{7/9}$, 39-dim)	ep+eo	Yes	Hub connector
S4 ($V_{2/3}$, 26-dim)	ep+co	Yes	Hub connector
S6 ($V_{5/9}$, 39-dim)	ep+eo	Yes	Primary hub (degree 5)
S7 ($V_{5/9}$, 66-dim)	cp+ep+co+eo	Yes	Secondary hub (degree 3)
S9 ($V_{1/3}$, 27-dim)	cp+co	Yes	CP-mediated channel to S8

S6 (2 pairs) and S7 (2 pairs) are the dominant T7 mediators; S4 and S9 each mediate 1 pair (see Rubik’s Cube Realization below). S1 is hybrid by block support (cp+ep) but transport-inert — a G -invariant subrepresentation with $K < 10^{-14}$ to all other sectors.

This definition makes precise the C2 condition in the T7 Theorem (see below): “transport-active hybrid projector” is now a fully defined mathematical object.

Conceptual note — hybrid sectors cannot be reduced to predominant blocks.

A common error in earlier computational analysis was to classify each sector by its *predominant* block — the block with largest trace contribution — and treat that as the sector’s block assignment. This is incorrect. Sectors S1 (cp+ep), S4 (ep+co), S7 (cp+ep+co+eo), and S9 (cp+co) each have *multi-block support* — their projectors have nonzero restriction to more than one invariant block. Collapsing a hybrid sector to its predominant block destroys the geometric feature that enables T7 morphisms: the projector’s ability to bridge two representation blocks in a single composition step. S7, for example, is split approximately 36%/39%/12%/13% across cp/ep/co/eo — calling it “an ep sector” loses its bridging role. The entire transport topology, M_2 overlap structure, and T7 mediation depend on **multi-block support**, not dominant-block assignment.

Lemma 2.3 (Schur-Locality: Why the Lie Algebra Cannot Cross Blocks).

The representation-level obstruction that underlies the entire paper is a structural consequence of Schur's lemma and the isotypic decomposition. This section makes it explicit.

Isotypic decomposition. Let $V = \bigoplus_{\tau \in \hat{G}} V_\tau$ where $V_\tau \cong M_\tau \otimes W_\tau$ is the τ -isotypic component: M_τ the irreducible module, W_τ the multiplicity space. By Schur's lemma:

$$\text{End}_G(V) = \bigoplus_{\tau} I_{M_\tau} \otimes \text{End}(W_\tau), \quad \text{Hom}_G(V_\tau, V_{\tau'}) = 0 \quad (\tau \neq \tau')$$

The double commutant theorem gives the image of the group algebra:

$$\mathcal{A} = \text{span}\{\rho(g) : g \in G\} = \text{End}_{\text{End}_G(V)}(V) = \bigoplus_{\tau} \text{End}(M_\tau) \otimes I_{W_\tau}$$

Block decomposition. The paper's working blocks (cp, ep, co, eo) are coarser than the isotypic decomposition: each block $V_b = \bigoplus_{\tau \in \text{Supp}(V_b)} V_\tau$ is a union of complete isotypic components. Consequently:

$$\rho(g) = \bigoplus_b \rho_b(g), \quad \mathcal{A} = \bigoplus_b \mathcal{A}_b$$

where $\mathcal{A}_b = \text{span}\{\rho_b(g)\}$ acts on V_b and vanishes on $V_{b' \neq b}$.

Proposition 2.4 (Schur-locality).

The matrix logarithm preserves the block-diagonal structure inherited from the isotypic decomposition:

$$A_g = \log \rho(g) = \bigoplus_b \log \rho_b(g) \in \mathcal{A}$$

Therefore the Lie algebra generated by the $\{A_g\}$ is contained in $\mathcal{A}$:

$$\mathcal{L} = \text{Lie}\{A_g : g \in G\} \subset \mathcal{A} = \bigoplus_b \mathcal{A}_b$$

Every Lie monomial at every depth — and by extension every element of the universal enveloping algebra $\mathcal{U}(\mathcal{L})$ — lives in $\mathcal{A}$ and is therefore block-diagonal. The Lie algebra cannot create new inter-block morphisms: this is a categorical obstruction, not a numerical one. $\text{Hom}_{\mathcal{L}}(V_A, V_B) = 0$ whenever $A \neq B$.

This is Layer A of the obstruction the T7 theorem builds on.

Concrete form (Isotypic Support Necessity). For computations with sector projectors, the same principle takes this form. Define the isotypic support $\text{Supp}(E_\alpha) = \{\tau \in \hat{G} : \text{Tr}(P_\alpha \Pi_\tau) > 0\}$, where Π_τ is the projector onto the τ -isotypic component of V .

If $\text{Supp}(E_\alpha) \cap \text{Supp}(E_\beta) = \emptyset$, then for all $g \in G$:

$$T_{\alpha\beta}(g) = P_\alpha \rho(g) P_\beta = 0$$

Proof. Take $v \in E_\beta$. Decompose: $v = \sum_{\tau \in \text{Supp}(E_\beta)} v_\tau$. By Schur's lemma, $\rho(g)$ preserves each isotypic component. P_α annihilates all $\tau \notin \text{Supp}(E_\alpha)$. Since the supports are disjoint, $P_\alpha \rho(g) v = 0$. $\square$

Corollary 2.5 (Composition can bridge disjoint support).

$P_\alpha \rho(g) P_\gamma \rho(h) P_\beta$ can be nonzero even when $\text{Supp}(E_\alpha) \cap \text{Supp}(E_\beta) = \emptyset$, provided an intermediate hybrid sector γ shares isotypic support with both: its V_A component receives $\rho(h)v$ and its V_B component passes the result to P_α .

This proves that **zero cross-block direct morphisms is a theorem, not an observation.**

3 Part I — The Accessibility Hierarchy

Remark 3.1 (Open question).

At what Lie-algebraic depth can sector α reach sector β ?

3.1 Level 0 — Direct Transport: K

$$K_{\alpha\beta} = \max_{g \in S} \|P_\alpha \rho(g) P_\beta\|_F$$

At 9-sector resolution (CCS-I §2.2): 20 directed edges (10 undirected pairs), **all block-preserving** — zero cross-block direct edges. S1 (V_1 , 20-dim, cp+ep) has $K < 10^{-14}$ with all other sectors — a G-invariant subrepresentation. S6 (39-dim, ep+eo, $V_5/9$) is the primary hub (degree 5). S7 (66-dim, all blocks, $V_5/9$) is the secondary hub (degree 3).

Structural Observation ([1], Obs. A). Nonempty Supp_nc intersection is sufficient for direct transport: $\text{Supp_nc}(\alpha) \cap \text{Supp_nc}(\beta) \neq \emptyset \implies K_{\alpha\beta} > 0$. All 9 Type I edges in the Rubik cube transport graph satisfy this criterion. The converse is an empirical regularity with a single exception: S8 (pure CP) $\leftrightarrow$ S9 (CP+CO) has $K = 2.83$ despite both having empty Supp_nc — the commutative CP-mediated Type II channel. Verified on the Rubik cube and S_3 negative controls; no first-principles derivation is claimed.

3.2 Level 1 — Gradient Transport: κ_0

$$\kappa_0(\alpha, \beta) = \max_{g \in S} \|P_\alpha A_g P_\beta\|_F$$

Computed over all 18 A_g . At 9-sector resolution — canonical matrix in (CCS-I §2.3). Gradient transport preserves all $K > 0$ channels: if $K_{\alpha\beta} > 0$ then $\kappa_0(\alpha, \beta) > 0$, symmetric to $< 1.6 \times 10^{-8}$. The 9-sector resolution reveals intra- $V_5/9$ structure invisible at 6-layer: S5 (1-dim, pure EO), S6 (39-dim, ep+eo, primary hub), and S7 (66-dim, all-block, secondary hub) are distinct sectors with distinct κ_0 profiles.

3.3 Level 2 — Curvature Transport: κ_1

$$\kappa_1(\alpha, \beta) = \max_{g, h \in S} \|P_\alpha [A_g, A_h] P_\beta\|_F$$

At 9-sector resolution — canonical matrix in (CCS-I §2.4). Figure 2 illustrates the geometric mechanism of curvature emergence at the Lie barrier.

Curvature Emergence: First-Order Frozen, Second-Order Accessible

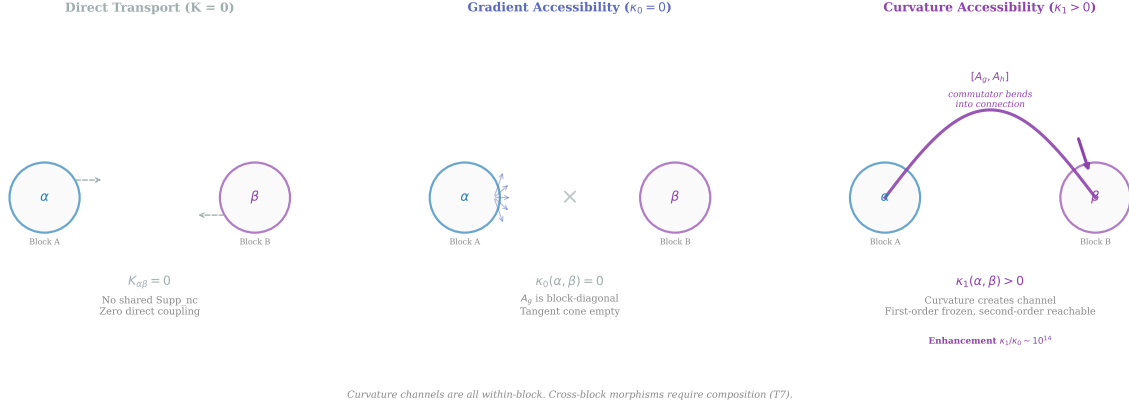


Figure 2. Geometric mechanism of curvature emergence at the Lie barrier: $\kappa_0 = 0$ (gradient freeze) on cross-block pairs means no single generator can bridge them at first order, while $\kappa_1 > 0$ (curvature active) within blocks means Lie brackets $\frac{1}{2}[QT^a, QT^b]$ produce second-order transport. All 7 curvature channels ($\kappa_1 > 0$) are confined within blocks, confirming that Lie brackets cannot cross block boundaries. Curvature and transport are independent structural dimensions — an edge can be gradient-only ($\kappa_0 > 0$, $\kappa_1 = 0$), curvature-assisted ($\kappa_0 > 0$, $\kappa_1 > 0$), or composition-only ($\kappa_0 = \kappa_1 = 0$).

Seven pure curvature channels at 9-sector — pairs with $\kappa_0 \approx 0$ but $\kappa_1 > 0$: $S2 \leftrightarrow S3$ ($\kappa_1=0.71$, shared EO), $S2 \leftrightarrow S7$ ($\kappa_1=1.01$, shared EO), $S3 \leftrightarrow S4$ ($\kappa_1=4.27$, shared EP), $S3 \leftrightarrow S5$ ($\kappa_1=1.01$, shared EO), $S4 \leftrightarrow S7$ ($\kappa_1=6.17$, shared EP+CO), $S5 \leftrightarrow S7$ ($\kappa_1=1.42$, shared EO), $S7 \leftrightarrow S8$ ($\kappa_1=6.98$, shared CP). The largest curvature-only channel at 9-sector is $S7 \leftrightarrow S8$ with $\kappa_0 \sim 10^{-14}$ and $\kappa_1 = 6.98$ (enhancement $\sim 5 \times 10^{14}$).

All 7 pure curvature channels are within-block — every pair shares at least one block (EP, EO, CO, or CP). Zero cross-block curvature channels at any resolution. **Lie curvature creates new within-block channels but cannot bridge blocks.** The 6-layer analysis identified 6 such channels; the 9-sector refinement resolves one more ($S2 \leftrightarrow S7$ distinguishing intra- $V_5/9$ EO coupling from the broader $S2 \leftrightarrow S6/S7$ pairing at 6-layer).

3.4 Higher Depth: κ_2 and Beyond

κ_2 (nested commutators $[[A_g, A_h], A_k]$) further amplifies all channels but creates no new cross-block ones. The pattern is stable: κ_d grows with d within blocks, stays zero across blocks. The Lie algebra is block-preserving at **all** depths (Lemma 4.1, see below).

3.5 Three Accessibility Classes

The κ hierarchy partitions sectors into three classes:

Class	Sectors	κ_0 (gradient)	κ_1 (curvature)	Mechanism
I	S1 (V_1)	0	0	G-invariant subrepresentation
II	Most non-S1 pairs	> 0	> 0	Gradient + curvature coupled
III	$S3 \leftrightarrow S4$, $S7 \leftrightarrow S8$, etc.	≈ 0	> 0	Pure curvature (Krawtchouk-order mismatch)

Class I is subrepresentation isolation — the sector is an actual G -subrepresentation, hence invariant under all $\rho(g)$ and A_g .

Class II is the generic case — individual Lie generators already couple the sectors, curvature amplifies.

Class III is the Krawtchouk-order mismatch mechanism: sectors in different Krawtchouk eigenspaces (e.g., $k = 2$ vs $k = 3$) that share block support. Individual A_g preserve Krawtchouk order (they live in the Bose-Mesner algebra); commutators $[A_g, A_h]$ mix across orders. The enhancement ratio $\kappa_1/\kappa_0 \sim 10^{14}$ is the signature of pure curvature coupling. At 9-sector resolution, 7 pure curvature channels are identified — all within-block, zero cross-block.

These three classes exhaust Lie-generated accessibility. What lies beyond them — compositional accessibility — is the subject of Parts II–III.

Connection to classical accessibility theory. The κ_d hierarchy mirrors the Lie bracket accessibility test in nonlinear control [2,3,4]: κ_0 corresponds to the Lie algebra generated by $\{A_g\}$, κ_1 to the first-order bracket $[A_g, A_h]$, and higher κ_d to iterated brackets. The key difference is that classical accessibility concerns the Lie algebra generated by vector fields on a manifold, while here the Lie generators A_g act linearly on the fixed representation space V . The block-preserving Lemma 4.1 (§4.1) is the finite-group analogue of the fact that the accessibility distribution respects invariant foliations [4, Ch.5].

4 Part II — Why Lie-Generated Accessibility Cannot Cross Blocks

Figure 3 illustrates the Lie barrier and the T7 bridge mechanism.

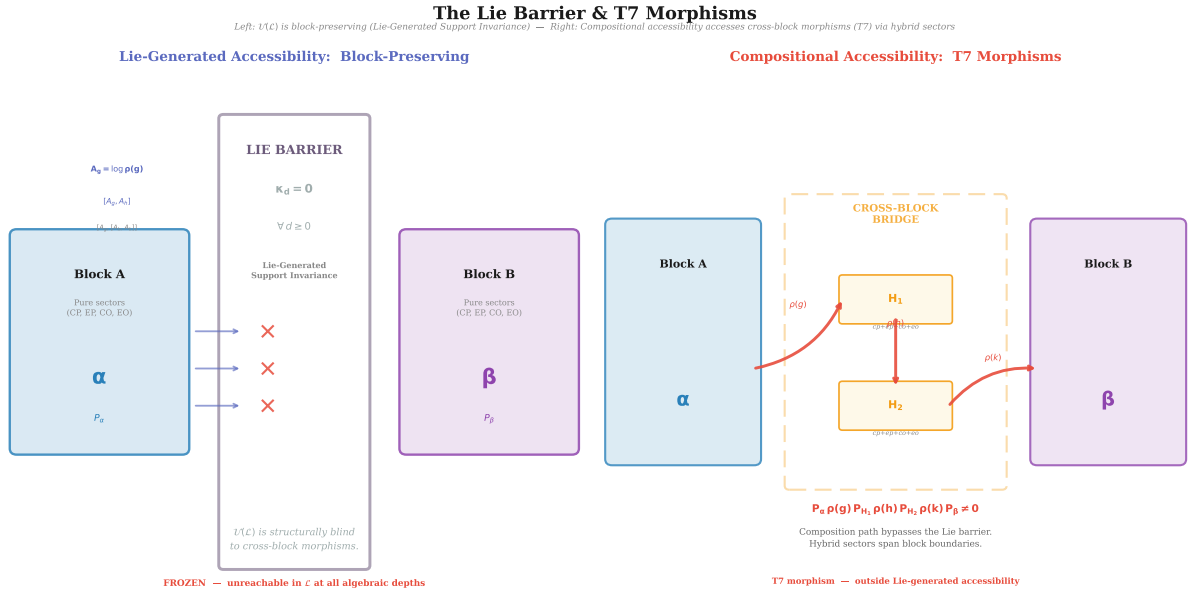


Figure 3. The Lie barrier: continuous tangent dynamics are confined within blocks (gradient κ_0 and curvature κ_1 cannot cross block boundaries), but compositional accessibility bridges blocks through hybrid sectors via length-2 paths. On the left, Lie-generated flows hit a wall at the block boundary and freeze; on the right, the T7 bridge bypasses the wall through a hybrid intermediate sector. Five T7 morphisms cross the block boundary, establishing that compositional accessibility strictly exceeds Lie-generated accessibility.

4.1 Lie-Generated Support Invariance

Layer A (Lemma 2.3) established containment in the group algebra: $\mathcal{L} \subset \mathcal{A} = \bigoplus_b \mathcal{A}_b$. Layer B extends this to the universal enveloping algebra — the full associative algebra generated by $\mathcal{L}$ under multiplication, not just the Lie subalgebra under commutator bracket.

Lemma 4.1 (Lie-Generated Support Invariance).

Let $V = V_A \oplus V_B$ be a decomposition into G -invariant blocks, and let $\rho(g) = \rho_A(g) \oplus \rho_B(g)$ for all $g \in G$. Define $\mathcal{L} = \text{Lie}\{A_g : g \in G\}$ where $A_g = \log \rho(g)$. If P_α and P_β are projectors with disjoint block support ($P_\alpha = P_\alpha \Pi_A$, $P_\beta = \Pi_B P_\beta$), then:

$$P_\alpha X P_\beta = 0 \quad \forall X \in \mathcal{U}(\mathcal{L})$$

where $\mathcal{U}(\mathcal{L})$ is the universal enveloping algebra of $\mathcal{L}$ — the associative algebra linearly spanned by finite products of the form $X_1 X_2 \cdots X_k$ for $X_i \in \mathcal{L}$.

Equivalently: if $P_\alpha \mathcal{L} P_\beta = \{0\}$, then $P_\alpha \mathcal{U}(\mathcal{L}) P_\beta = \{0\}$. Lie-generated accessibility preserves block support at all algebraic depths.

Proof. For each g , $A_g = \log \rho(g) = \log \rho_A(g) \oplus \log \rho_B(g)$ — the matrix logarithm preserves block structure. The commutator of block-diagonal matrices is block-diagonal:

$$[X_A \oplus X_B, Y_A \oplus Y_B] = [X_A, Y_A] \oplus [X_B, Y_B]$$

so $\mathcal{L} \subset \mathcal{A}_A \oplus \mathcal{A}_B$ (all Lie generators and their commutators are block-diagonal). Now any $X \in \mathcal{U}(\mathcal{L})$ is a finite linear combination of finite products $Y_1 Y_2 \cdots Y_k$ with $Y_i \in \mathcal{L}$. Each Y_i is block-diagonal. The product of block-diagonal matrices is block-diagonal:

$$(C_A \oplus C_B)(D_A \oplus D_B) = C_A D_A \oplus C_B D_B$$

By induction on product length, every such product — and hence every element of $\mathcal{U}(\mathcal{L})$ — is block-diagonal. Projecting onto disjoint blocks annihilates:

$$P_\alpha X P_\beta = P_\alpha \Pi_A X \Pi_B P_\beta = P_\alpha \cdot 0 \cdot P_\beta = 0$$

□

This is the central structural fact of the paper. The universal enveloping algebra $\mathcal{U}(\mathcal{L})$ — the totality of Lie-generated accessibility — is block-preserving. No polynomial in the Lie generators, of any degree or depth, can couple sectors with disjoint block support.

Remark 4.2 (Why $\mathcal{U}(\mathcal{L})$ and not just $\mathcal{L}$).

The Lie algebra $\mathcal{L}$ controls commutator-depth accessibility (κ_d), but finite-time Lie-generated motion is not limited to single commutators. Products of exponentials $\exp(t_1 A_{g_1}) \exp(t_2 A_{g_2}) \cdots$ expand, via the Baker-Campbell-Hausdorff formula, into polynomials in the generators — elements of the universal enveloping algebra $\mathcal{U}(\mathcal{L})$, the associative algebra spanned by finite products $X_1 \cdots X_k$ with $X_i \in \mathcal{L}$. Proving $\mathcal{U}(\mathcal{L})$ is block-preserving (Lemma 4.1) therefore proves that no Lie-generated motion — infinitesimal or finite, at any order — can couple sectors with disjoint block support. Lemma 4.1 is strictly stronger than $\kappa_d = 0$ at every depth d .

Remark 4.3 (Terminology — Lie-generated vs compositional).

The name **Lie-Generated Support Invariance** reflects what the lemma establishes: Lie-generated accessibility preserves block support. The term “curvature” is retained for κ_1 specifically — the commutator-mediated channel within Lie-generated accessibility — but Lemma 4.1 governs the entire Lie-generated structure, not curvature alone.

4.2 Cross-Block Lie Inaccessibility

For a pure-A sector α (projector supported entirely in V_A) and a pure-B sector β (supported entirely in V_B), and any block-diagonal operator $C = C_A \oplus C_B$:

$$P_\alpha C P_\beta = (P_\alpha \Pi_A)(C_A \oplus C_B)(\Pi_B P_\beta) = P_\alpha \cdot (\Pi_A(C_A \oplus C_B)\Pi_B) \cdot P_\beta = P_\alpha \cdot 0 \cdot P_\beta = 0$$

By Lemma 4.1, every C_d at any Lie depth d is block-diagonal. Therefore:

$$\kappa_d(\alpha, \beta) = 0 \quad \forall d \geq 0$$

Cross-block Lie-generated accessibility is identically zero at all depths. The continuous limit, no matter how many commutator levels are included, cannot couple sectors supported in disjoint blocks.

4.3 Within-Block Curvature Channels

In contrast, **within** a block, commutators create genuinely new transport. The 7 pure curvature channels ($\kappa_0 \approx 0$, $\kappa_1 > 0$) all share block support. The Krawtchouk-order mixing mechanism operates entirely within the cp/EP blocks — it cannot cross the block boundary.

This separation is absolute:

- **Within-block:** Lie gradient + Lie curvature + higher depth — rich Lie-generated accessibility
- **Cross-block:** identically zero at all Lie depths — compositional accessibility is the sole channel

4.4 Formal Definitions

Before the T7 theorem, we define the two accessibility structures the theorem compares.

Definition 4.4 (Compositional accessibility).

Let $\{E_\alpha\}$ be the primitive sectors. A morphism $E_\beta \rightarrow E_\alpha$ exists in **compositional accessibility** if there exists a finite sequence of intermediate sectors $\gamma_1, \dots, \gamma_{n-1}$ and group elements $g_1, \dots, g_n \in G$ such that:

$$P_\alpha \rho(g_n) P_{\gamma_{n-1}} \cdots P_{\gamma_1} \rho(g_1) P_\beta \neq 0$$

This is the accessibility structure of the discrete group action: which sectors can exchange amplitude through any finite sequence of generators. Length-1 morphisms are the direct morphisms $T_{\alpha\beta}(g) = P_\alpha \rho(g) P_\beta$ ([1]); the closure under finite composition is denoted $\overline{\mathcal{T}}$.

Definition 4.5 (Lie-generated accessibility).

Lie-generated accessibility has the same objects $\{E_\alpha\}$ but its morphisms are restricted to those realizable by elements of the universal enveloping algebra: a morphism $E_\beta \rightarrow E_\alpha$ exists when $P_\alpha X P_\beta \neq 0$ for some $X \in \mathcal{U}(\mathcal{L})$, where $\mathcal{L} = \text{Lie}\{A_g\}$. This is the accessibility structure of the continuous (Lie) limit.

Containment. Lie-generated accessibility is contained in compositional accessibility ($\mathcal{L} \subset \overline{\mathcal{T}}$), since every Lie-generated morphism is a limit of compositional morphisms via the exponential map. Lemma 4.1 (§4.1) establishes that Lie-generated accessibility is block-preserving: no morphism connects sectors with disjoint block support. The central question of §5 is whether this containment is strict — whether compositional accessibility contains cross-block morphisms that Lie-generated accessibility lacks.

5 Part III — The T7 Morphism Theorem

This is the climactic theorem of the trilogy.

Definition 5.1 (T7 Morphism).

A **T7 morphism** (for “type 7,” the seventh structural pattern identified in the Rubik’s cube spectral enumeration, following six within-block accessibility types) between sectors (α, β) satisfies three conditions:

1. $K_{\alpha\beta} = 0$ — no direct morphism (single generator)
2. $\kappa_d(\alpha, \beta) = 0$ **for all** d — no Lie-generated accessibility at any depth
3. $\exists \gamma$ **with** $K_{\alpha\gamma} > 0$ **and** $K_{\gamma\beta} > 0$ — reachable via length-2 discrete composition through a hybrid sector

The morphism is cross-block and outside Lie-generated accessibility: discrete group action can bridge it through a hybrid sector; the continuous Lie limit cannot — at any order of commutators, at any polynomial degree in $\mathcal{U}(\mathcal{L})$.

Structural detection vs numerical κ threshold. The canonical T7 detection criterion is *structural*: condition 2 ($\kappa_d = 0$ for all d) is verified via block-set disjointness — $\text{block support}(\alpha) \cap \text{block support}(\beta) = \emptyset$ implies $\kappa_d(\alpha, \beta) = 0$ for all d as an exact consequence of Lemma 4.1 (Lie-Generated Support Invariance). This is a representation-theoretic obstruction test, not a floating-point heuristic. κ_d numerical thresholds ($\text{TOL_KAPPA} = 10^{-6}$) are used only as consistency sanity checks — they confirm that the computed κ values are consistent with the structural prediction, but the prediction itself is derived from block algebra, not from comparing floating-point norms to a tolerance. Earlier experimental versions used purely numerical κ thresholds ($\text{TOL} = 10^{-10}$), which incorrectly removed the $S4 \leftrightarrow S5$ T7 channel due to logm numerical leakage ($\sim 10^{-8}$). The present canonical definition is set-based and representation-structural.

5.1 T7 Theorem (Formal Statement)

Setup. Let:

- **C0 (Center Incompleteness).** $Z \subsetneq \text{Comm}(\rho)$ ([1], Notation Table). The sector-decomposing center Z is a proper subalgebra of the full commutant $\text{Comm}(\rho)$. When $Z = \text{Comm}(\rho)$, sectors are G -invariant, K is purely diagonal, and T7 is structurally impossible. C0 is the foundational condition: without center incompleteness, C1–C3 cannot produce T7.
- G be a finite group
- (V, ρ) a finite-dimensional unitary representation: $\rho(g) = \rho_A(g) \oplus \rho_B(g)$, $V = V_A \oplus V_B$
- $S \subset G$ an inverse-closed generator subset
- $Z = \langle A_{S_1}, \dots, A_{S_k} \rangle$ a commutative subalgebra of the averaging algebra ($k \geq 1$), containing A_S , where each $A_{S_i} = \frac{1}{|S_i|} \sum_{g \in S_i} \rho(g)$ is the averaging operator over a generator subset $S_i \subseteq S$
- $\{E_\alpha\}$ the primitive sectors from joint diagonalization of Z
- A sector E_α is **pure-A** if $P_\alpha V_B = \{0\}$, **pure-B** if $P_\alpha V_A = \{0\}$, **hybrid** otherwise

Structural Conditions.

(C1) Shared noncommutative support. There exists an irreducible representation τ of G appearing in both ρ_A and ρ_B with multiplicity ≥ 1 in each.

Equivalently: $\text{Supp}_{\text{nc}}(V_A) \cap \text{Supp}_{\text{nc}}(V_B) \neq \emptyset$.

(C2) Transport-active hybrid projector. There exists a primitive sector E_γ (a joint eigenspace of Z) whose projector P_γ satisfies both:

- (i) **Hybrid block support:** $P_\gamma V_A \neq \{0\}$ and $P_\gamma V_B \neq \{0\}$ — the projector spans both blocks.
- (ii) **Transport-active (computable criterion):** Define the isotypic support $\text{Supp}(E_\alpha) = \{\tau \in \hat{G} : \text{Tr}(P_\alpha \Pi_\tau) > 0\}$ where Π_τ is the projector onto the τ -isotypic component of V . Then E_γ is *transport-active* if and only if:

$$\text{Supp}(E_\alpha) \cap \text{Supp}(E_\gamma) \neq \emptyset \quad \text{and} \quad \text{Supp}(E_\gamma) \cap \text{Supp}(E_\beta) \neq \emptyset$$

for some pure-A sector α and pure-B sector β . Equivalently, there exists an irreducible representation τ common to all three isotypic supports, and within each block the τ -multiplicity spaces of P_α and P_γ (in V_A) and of P_γ and P_β (in V_B) have nonzero overlap under $\rho(g)$. This is computable: given the isotypic projectors Π_τ — obtained by diagonalizing the full commutant algebra $\text{Comm}(\rho(G))$, which for this representation is a 610-dimensional commutative algebra computed via index-pair orbit decomposition (see `experiments/paper1/isotypic_decomposition.py`) — verify $\text{Tr}(P_\alpha \Pi_\tau) > 0$, $\text{Tr}(P_\gamma \Pi_\tau) > 0$, $\text{Tr}(P_\beta \Pi_\tau) > 0$ for at least one $\tau \in \hat{G}$. For the S_3 negative controls, the isotypic decomposition is explicit in (CCS Appendix G).

A *spectral hybrid* — a joint eigenspace spanning both blocks but arising from eigenvalue coincidence in the commutative center Z without shared τ -multiplicity geometry — satisfies (i) but not (ii). Such sectors are transport-inert: they provide no composition path, as confirmed by the S_3 inert hybrid counterexample (Appendix C).

(C3) Block-preserving dynamics. $\rho(g) = \rho_A(g) \oplus \rho_B(g)$ for all $g \in G$.

Theorem 5.2 (T7 Theorem).

Under C0–C3:

(a) Lie inaccessibility. For any pure-A sector α and pure-B sector β :

$$\kappa_d(\alpha, \beta) = 0 \quad \forall d \geq 0$$

No finite-depth Lie bracket sequence can couple them.

(b) Compositional accessibility. There exists a transport-active hybrid sector γ and group elements $g, h \in G$ such that:

$$P_\alpha \rho(g) P_\gamma \neq 0 \quad \text{and} \quad P_\gamma \rho(h) P_\beta \neq 0$$

Hence the composition path $\alpha \rightarrow \gamma \rightarrow \beta$ is accessible.

(c) Compositional accessibility $\supsetneq$ Lie-generated accessibility.

Proof. The proof is structured as a chain of three layers, each relying on the previous one. Layer A (Schur-locality, Lemma 2.3) establishes that the Lie algebra is contained in the block-diagonal subalgebra. Layer B (Support Invariance, Lemma 4.1) extends this to the full universal enveloping algebra — the totality of Lie-generated accessibility. Layer C (T7) shows that compositional accessibility circumvents this containment via hybrid sectors.

Layer A — Schur-locality (Lemma 2.3). The isotypic decomposition $V = \bigoplus_\tau M_\tau \otimes W_\tau$ gives $\rho(g) \in \mathcal{A} = \bigoplus_\tau \text{End}(M_\tau) \otimes I_{W_\tau}$. Since the blocks V_A, V_B are unions of complete isotypic components, $\mathcal{A} = \mathcal{A}_A \oplus \mathcal{A}_B$. The matrix logarithm preserves this structure: $A_g = \log \rho(g) \in \mathcal{A}$. Therefore:

$$\mathcal{L} = \text{Lie}\{A_g\} \subset \mathcal{A} = \mathcal{A}_A \oplus \mathcal{A}_B$$

Every Lie generator and every Lie bracket lives in the block-diagonal algebra. $\text{Hom}_{\mathcal{L}}(V_A, V_B) = 0$ — no Lie monomial at any depth can create an inter-block morphism. This is a categorical obstruction, not a numerical one.

Layer B — Lie-Generated Support Invariance (Lemma 4.1). Layer A shows $\mathcal{L} \subset \mathcal{A}_A \oplus \mathcal{A}_B$. Lemma 4.1 extends this to the universal enveloping algebra: every $X \in \mathcal{U}(\mathcal{L})$ is block-diagonal, because $\mathcal{U}(\mathcal{L})$ is linearly spanned by finite products of elements of $\mathcal{L}$, and the product of block-diagonal matrices is block-diagonal. Consequently:

$$P_\alpha X P_\beta = 0 \quad \forall X \in \mathcal{U}(\mathcal{L})$$

whenever α is pure-A and β is pure-B. This is strictly stronger than $\kappa_d = 0$ for all d : it says Lie-generated accessibility — not just the commutator-depth hierarchy — is block-preserving. Any channel between pure-A and pure-B objects is absent from $\mathcal{U}(\mathcal{L})$ entirely.

(a) Cross-block $\kappa_{\mathbf{d}} = 0$. Immediate from Layer B. Let α be pure-A ($P_\alpha V_B = \{0\}$), β pure-B ($P_\beta V_A = \{0\}$). For any block-diagonal $C = C_A \oplus C_B$:

$$P_\alpha C P_\beta = P_\alpha \Pi_A (C_A \oplus C_B) \Pi_B P_\beta = P_\alpha \cdot 0 \cdot P_\beta = 0$$

By Layers A+B, every $C \in \mathcal{U}(\mathcal{L})$ is block-diagonal. Hence $\kappa_d(\alpha, \beta) = 0$ for all d .

Layer C — Compositional accessibility circumvents the Lie obstruction. Layers A+B establish that Lie-generated accessibility is block-preserving: no Lie-generated morphism connects pure-A to pure-B. But the discrete group action $\rho(g)$ operates at the level of the representation itself, not at the level of $\mathcal{U}(\mathcal{L})$. A two-step composition through a hybrid projector P_γ is not an element of $\mathcal{U}(\mathcal{L})$ — it is a projector-mediated morphism of the form:

$$P_\alpha \rho(g) P_\gamma \rho(h) P_\beta$$

This object lives in compositional accessibility, not in Lie-generated accessibility.

(b) Discrete composition path exists. By C1, there exists a shared irreducible representation τ appearing in both ρ_A and ρ_B . By C2, there exists a transport-active hybrid sector γ with $\tau \in \text{Supp}(E_\gamma)$ and nonzero $K_{\alpha\gamma}$ and $K_{\gamma\beta}$ (the computable isotypic-support criterion of §5.2). By the concrete form of Lemma 2.3, $\text{Supp}(E_\alpha) \cap \text{Supp}(E_\gamma) \neq \emptyset$ (both carry τ in V_A) and $\text{Supp}(E_\gamma) \cap \text{Supp}(E_\beta) \neq \emptyset$ (both carry τ in V_B). Hence there exist $g, h \in G$ with $T_{\alpha\gamma}(g) = P_\alpha \rho(g) P_\gamma \neq 0$ and $T_{\gamma\beta}(h) = P_\gamma \rho(h) P_\beta \neq 0$. The composition path $\alpha \rightarrow \gamma \rightarrow \beta$ is accessible.

(c) Structural separation. Parts (a) and (b) together establish that Lie-generated accessibility is strictly contained in compositional accessibility:

$$\text{Lie-generated accessibility}|_{\text{cross-block}} = \emptyset, \quad \text{compositional accessibility}|_{\text{cross-block}} \neq \emptyset$$

Lie-generated accessibility has zero cross-block morphisms; compositional accessibility has nonzero cross-block morphisms. The T7 morphisms are precisely the cross-block morphisms outside Lie-generated accessibility. This is the central structural fact of the trilogy.

Why the Lie obstruction cannot be circumvented by higher commutators. A natural question: could a sufficiently high commutator depth d produce a cross-block channel? Layers A+B answer definitively: no. The obstruction is not that commutators fail to mix blocks at low depth — it is that the entire algebra $\mathcal{U}(\mathcal{L})$ is block-preserving. Lie-generated accessibility has no cross-block morphisms at any algebraic depth, because it lives inside $\mathcal{A}_A \oplus \mathcal{A}_B$ by the structural consequence of Schur's lemma (Layer A). Composition through a hybrid projector is a qualitatively different mathematical operation from Lie-polynomial generation. This is the structural separation that T7 names. $\square$

5.2 Necessity

Condition	Structural Role	Necessity Status
C1 Shared non-commutative support	Algebraic substrate — bridgeability (2.3)	Conjectured for general case; proved for abelian groups + isotypic representations; exhaustive search passed on all small-group systems tested (see below)
C2 Transport-active hybrid sector	Structural bridge — composition path exists	Definitional — follows from the computable criterion (see the T7 Theorem below); the distinction from inert spectral hybrids is verified
C3 Block-preserving dynamics	Dynamic separation — Lie inaccessibility	Proved (contrapositive of T7 definition; Lemma 4.1 guarantees all Lie monomials are block-diagonal)

N1 (C1 necessity — why the general proof is hard). Without shared noncommutative support, C1 asserts that no T7 morphism exists. In the abelian case, all irreps are 1-dimensional; the center Z acts as simultaneous eigenvalues on each irrep, making the sector partition a refinement of the irrep decomposition. Disjoint irreps yield sectors with disjoint isotypic support — Lemma 2.3 then forces $K=0$, blocking the composition path. For the isotypic case (both blocks carry only one irrep type), the shared support is automatic.

The difficulty in the general non-abelian case is that the center Z can mix distinct irreps within the same eigenspace (eigenvalue coincidence), producing spectral hybrids that carry disjoint irreps — a “not shared” scenario that could, in principle, circumvent C1. The question is whether such a spectral hybrid can ever be transport-active. All small-group systems tested to date — S_3 (all representation pairs: $\text{nat} \oplus \text{reg}$, $\text{reg} \oplus \text{reg}$, $\text{nat} \oplus \text{nat}$, $\text{std} \oplus \text{sign}$, etc.), Q_3 (quaternion, 8 elements), S_4 (24 elements), and $Z_2 \times Z_2$ — satisfy C1 necessity: no T7 morphism arises without a shared irrep. The S_3 “false T7” (standard vs. $\text{trivial} \oplus \text{sign}$) confirms the mechanism: eigenvalue coincidence creates a hybrid projector spanning both blocks, but without a shared irrep the hybrid fails the computable C2 criterion ($\text{Tr}(P_\alpha \Pi_\tau) = 0$ for all τ common to α and β).

N2 (C2 necessity). If no transport-active hybrid sector exists between α and β , no length-2 composition path through any mediating sector γ can carry amplitude between the two blocks. The contrapositive is built into the definition of C2: a hybrid P_γ is transport-active iff $K_{\alpha\gamma} > 0$ and $K_{\gamma\beta} > 0$ — exactly the third T7 condition. C2 necessity is therefore a definitional restatement of the composition-path requirement and does not require separate proof.

N3 (C3 necessity). If cross-block $\kappa_d(\alpha, \beta) > 0$ for some d , the pair is Lie-accessible and not a T7 morphism. The contrapositive is the T7 definition. Block-diagonal ρ is the natural sufficient condition; it is conjectured necessary for *systematic* cross-block T7 (a single non-block-diagonal generator would provide a Lie-accessible cross-block channel).

The C0–C3 chain is established for the Rubik cube (verified in §5.3). C2 and C3 are proved necessary (§5.4). C1 necessity is established for abelian and isotypic cases and supported by exhaustive search on all small-group systems tested; general C1 necessity is conjectural. A proof for general non-abelian groups would require showing that eigenvalue coincidence across disjoint irreps (in the commutative center Z) can never produce a transport-active hybrid sector — i.e., that the computable C2 criterion cannot be satisfied without shared isotypic support.

5.3 T7 and M_2 Are Independent Obstruction Types

The T7 mechanism is often conflated with the M_2 principle ([1] — noncommutative AW simple components as transport atoms). They are logically distinct:

Property	M_2 Obstruction	T7 Obstruction
Origin	Noncommutative AW components ($n_i \geq 2$)	Block-diagonal ρ + shared irrep + hybrid sector
Effect	Refinement failure, curvature transport	Cross-block T7 morphisms
Lie consequence	$\kappa_1 > 0$ within shared M_2	$\kappa_d = 0$ across blocks
Minimal system	Any noncommutative AW component	Rubik cube (228-dim, both T7 and M_2)
Within/cross block	Within-block	Cross-block

Co-occurrence: The Rubik cube exhibits both T7 (5 cross-block pairs) and M_2 curvature (7 within-block channels). The structural orthogonality is observed at scale: T7 morphisms are exclusively cross-block, curvature channels exclusively within-block. Neither S_3 negative control exhibits T7.

6 Part IV — S_3 Prototypes and Rubik Realization

Figure 4 shows the S_3 nat $\oplus$ reg prototype as a negative control for the C0–C3 conditions.

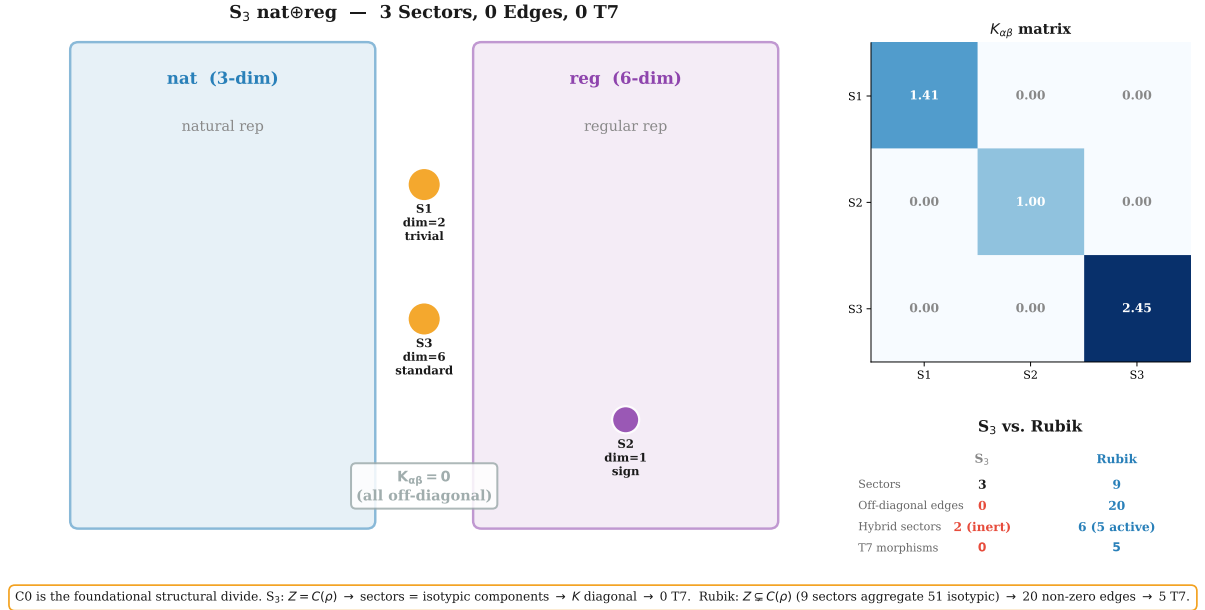


Figure 4. The S_3 nat $\oplus$ reg prototype (9-dimensional) as a negative control: 3 sectors from the canonical Center decomposition, 2 are hybrid, but C0 fails (Z coincides with isotypic components, K is diagonal), producing 0 T7 morphisms. Despite having hybrid sectors (S1, S3 span nat+reg), they are transport-inert — no cross-sector coupling exists. This negative control demonstrates that hybrid sectors alone are insufficient for T7: the center must be sufficiently incomplete (C0) for compositional accessibility to exceed Lie accessibility.

6.1 S_3 nat $\oplus$ reg — 9-dim (Negative Result)

$G = S_3$ (order 6), $\rho = \rho_{\text{nat}} \oplus \rho_{\text{reg}}$ (3-dim natural + 6-dim regular), S = all 6 group elements, $Z = \langle A_{\text{full}}, A_{\text{trans}} \rangle$.

Canonical decomposition. Center $\{A_{\text{full}}, A_{\text{trans}}\}$ yields 3 sectors (not 5 as previously reported; the earlier result was an artifact of a bug in the joint diagonalization that used standard basis indices instead of eigenvectors, fixed 2026-05-26).

Sector	dim	$\lambda(A_{\text{full}})$	$\lambda(A_{\text{trans}})$	Block support	Type
S1	2	1.0	1.0	nat+reg	hybrid
S2	1	0.0	-1.0	reg	pure
S3	6	0.0	0.0	nat+reg	hybrid

0 T7 morphisms. All cross-sector transport $K_{ij} = 0$ for $i \neq j$ ($K_{11} = 1.41$, $K_{22} = 1.0$, $K_{33} = 2.45$ are self-transport only). The hybrid sectors S1 and S3 are transport-inert: despite spanning both blocks, they have zero transport coupling to any other sector. C2 (transport-active hybrid sector) fails.

Structural lesson. A non-abelian group with block-diagonal representation is INSUFFICIENT for T7. The specific irrep structure must produce transport-active hybrid sectors — sectors that both span multiple blocks AND couple to sectors in different blocks via $K_{\alpha\gamma} > 0$. S_3 nat+reg has hybrid sectors but they are inert. The Rubik cube (228-dim) is the canonical realization satisfying all four C0–C3 conditions (5 T7 pairs, 9 sectors).

6.2 S_3 reg+reg — 12-dim (Negative Result)

$G = S_3$, $\rho = \rho_{\text{reg}} \oplus \rho_{\text{reg}}$, $S = \{3 \text{ transpositions}\}$, $Z = \langle A_{\text{full}}, A_{\text{trans}} \rangle$.

Canonical decomposition. Center $\{A_{\text{full}}, A_{\text{trans}}\}$ yields 3 sectors, ALL hybrid (span both reg copies). Zero pure sectors — no cross-block pure-sector pairs can exist. 0 T7 morphisms.

Sector	dim	$\lambda(A_{\text{full}})$	$\lambda(A_{\text{trans}})$	Block support
S1	2	1.0	1.0	A+B
S2	2	0.0	-1.0	A+B
S3	8	0.0	0.0	A+B

Structural lesson. When the center decomposition produces only hybrid sectors, there are no pure-block sectors to form cross-block T7 pairs. C0–C3 are characterizing conditions — their absence perfectly predicts T7 absence. The Rubik cube satisfies all four conditions; the S_3 canonical decomposition fails C0 ($Z = \text{Comm}(\rho)$), and neither S_3 negative control satisfies C2.

6.3 Rubik’s Cube — 228-dim, Same Mechanism at Scale

The Rubik’s cube realizes the identical C0–C3 mechanism.

C1 — Shared noncommutative support. The EP block (144-dim, $\| [QT^0, QT^1] \| = 2.74$) carries the $M_2(\mathbb{C})^4$ noncommutative AW components. Multiple sectors share EP support (S3, S4, S6, S7), with M_2 components distributed across them.

C2 — Hybrid projectors. S6 (ep+eo, 39-dim, k=4) and S7 (cp+ep+co+eo, 66-dim, k=4) span multiple blocks. S6 is the primary hub (degree 5), S7 the secondary (degree 3). Both are transport-active — they route composition paths between pure-EP, pure-EO, pure-CP, and pure-CO sectors.

C3 — Block-preserving dynamics. $\rho = \rho_{\text{cp}} \oplus \rho_{\text{ep}} \oplus \rho_{\text{co}} \oplus \rho_{\text{eo}}$ is block-diagonal by construction. Lemma 4.1 applies.

5 T7 morphisms (CCS-I §2.5):

Pair	Block support	Mediation path
$S2(eo) \leftrightarrow S4(ep+co)$	$eo \cap (ep+co) = \emptyset$	$S2 \rightarrow S6 \rightarrow S4$
$S3(ep+eo) \leftrightarrow S9(cp+co)$	$(ep+eo) \cap (cp+co) = \emptyset$	$S3 \rightarrow S7 \rightarrow S9$
$S4(ep+co) \leftrightarrow S5(eo)$	$(ep+co) \cap eo = \emptyset$	$S4 \rightarrow S6 \rightarrow S5$
$S4(ep+co) \leftrightarrow S8(cp)$	$(ep+co) \cap cp = \emptyset$	$S4 \rightarrow S9 \rightarrow S8$
$S6(ep+eo) \leftrightarrow S9(cp+co)$	$(ep+eo) \cap (cp+co) = \emptyset$	$S6 \rightarrow S7 \rightarrow S9$ (also $S6 \rightarrow S4 \rightarrow S9$)

All 5 T7 morphisms are **cross-block** (disjoint block support). All are mediated through the S6–S7–S9 hub complex (canonical mediation statistics: S6:2, S7:2, S9:1). Zero within-block T7 morphisms.

7 curvature channels — all **within-block** (share at least one block). Zero cross-block curvature channels. Curvature is block-preserving at scale, exactly as the S_3 negative controls predict.

S1 is NOT T7. S1 (V_1 , 20-dim, cp+ep) has $K = \kappa_0 = \kappa_1 = 0$ with all other sectors, but it is a genuine G -invariant subrepresentation — no composition path exists. This is subrepresentation isolation, not T7. Figure 5 shows the hybrid bridge topology at full 9-sector resolution.

6.4 Scale Comparison

Property	S_3 nat $\oplus$ reg	S_3 reg $\oplus$ reg	Rubik
Dimension	9	12	228
Sectors	3	3	9
Blocks	2 (nat, reg)	2 (reg, reg)	4 (CP, EP, CO, EO)
Hybrid sectors	2 (both inert)	3 (all hybrid)	6 (5 transport-active)
Curvature ($\kappa_1 > 0$)	0	0	7 (within-block)
T7 morphisms	0	0	5 (cross-block)
C2 satisfied?	No	No	Yes
M_2 present?	Yes (reg: $M_2(C)$)	Yes	Yes ($A_{EP} \cong M_2^4 \oplus M_1^4$)

The S_3 negative controls (9-dim, 12-dim) demonstrate that C0–C3 are non-trivial: S_3 canonical fails C0 (center completeness forces diagonal K), and neither negative control satisfies C2 (transport-active hybrid), despite having non-abelian group structure, block-diagonal representations, and shared irreps. The specific irrep structure must also produce transport-active hybrid sectors.

Cross-Block Bridge Topology — Composition-Only Morphisms Exist ONLY Through Hybrids

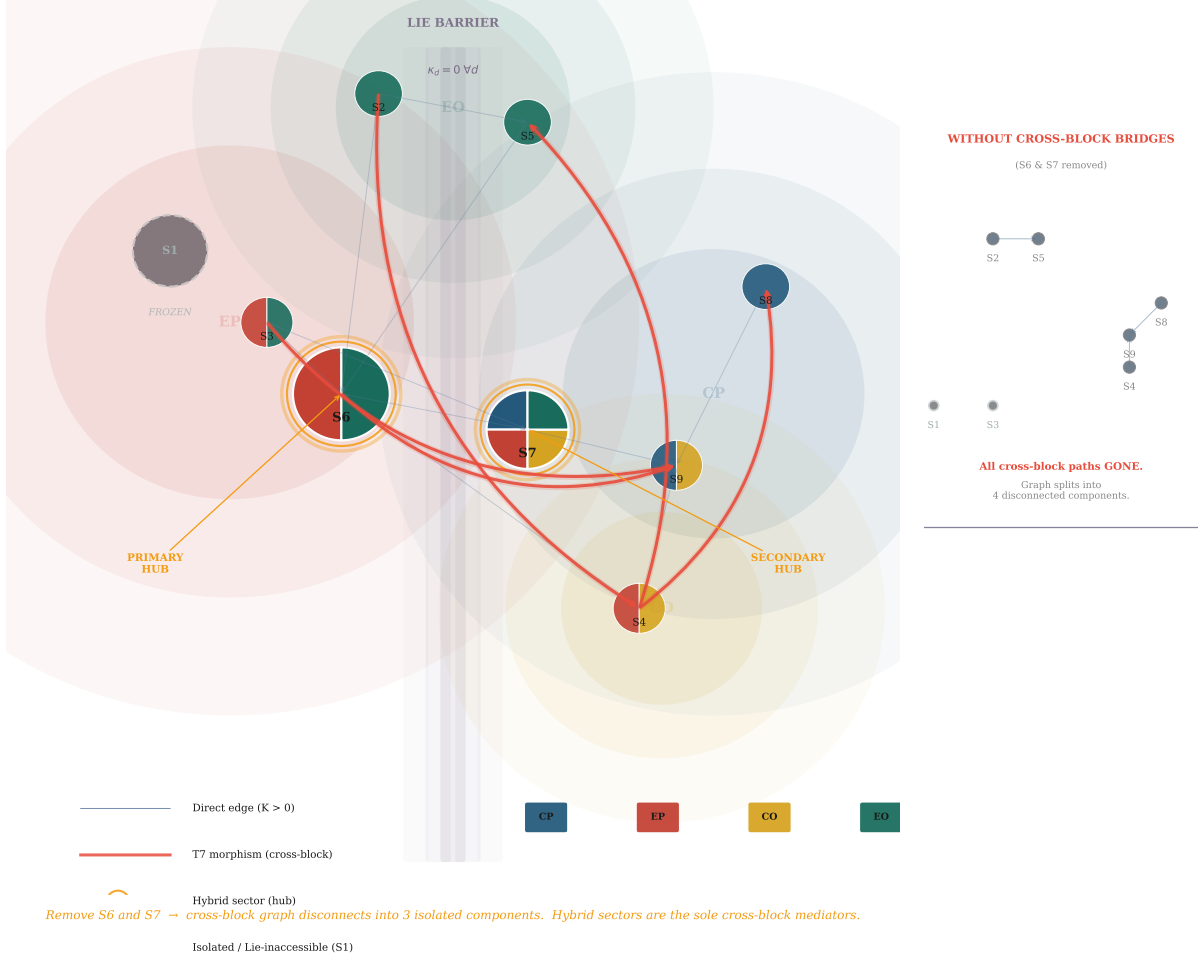


Figure 5. Hybrid bridge topology at 9-sector resolution: the four invariant blocks (CP, EP, CO, EO) partition sectors by support, with hybrid sectors (supported on multiple blocks) acting as bridges across the block boundary. All five T7 cross-block pairs involve at least one hybrid intermediate; pure sectors within a single block cannot reach sectors in different blocks through Lie-generated paths. Hybrid sectors are the structural precondition for compositional accessibility to exceed Lie accessibility.

7 Part V — The Structure Behind T7: Lie vs. Composition

The preceding sections established the T7 phenomenon at three scales. This section steps back and asks: **what kind of mathematical structure does T7 reveal?**

The answer is a strict separation between two accessibility structures on the same set of sectors. Lie-generated accessibility is block-preserving (Layers A+B). Compositional accessibility is not. The gap between them — exactly the T7 morphisms — is the structural signature of a fundamental mismatch between infinitesimal generation and discrete composition in finite group representations.

7.1 Two Generation Principles

The paper’s data organizes around a central dichotomy — not in the group, not in the representation, but in **how accessibility is generated**:

	Lie-generated accessibility	Compositional accessibility
Generators	$A_g = \log \rho(g)$	$\rho(g)$
Operation	Lie bracket $[A_g, A_h]$, universal enveloping algebra $\mathcal{U}(\mathcal{L})$	Concatenation through sector projectors P_γ
Geometry probed	Tangent space of group orbit (infinitesimal)	Global projector geometry (finite)
Block crossing	Forbidden — $\mathcal{L} \subset \bigoplus_b \mathcal{A}_b$ (Layers A+B)	Enabled — through hybrid sectors (C2)
Increasing reach	Commutator depth ($d = 0, 1, 2, \dots$), polynomial degree	Path length ($n = 1, 2, \dots$)
Structural scope	Inside blocks — $\mathcal{U}(\mathcal{L})$ is block-preserving	Across blocks — projector-mediated morphisms

The continuous limit — the standard mathematical tool for smooth accessibility analysis — replaces $\rho(g)$ with its logarithm A_g . This collapses projector-mediated cross-block paths into block-diagonal operators. Lie-generated accessibility is structurally blind to cross-block composition. **Compositional accessibility restores the morphisms that the continuous limit annihilates.**

This is not a failure of the Lie algebra — Lie-generated accessibility is perfectly adequate for within-block morphisms. It is a structural mismatch between two generation principles, neither fully reducible to the other.

7.2 The Structural Separation

Lie-generated accessibility is contained in compositional accessibility. Any Lie monomial $C \in \mathcal{U}(\mathcal{L})$ can be expressed via the exponential map as a limit of group compositions, so every Lie-generated morphism is a compositional morphism. But the containment is strict: compositional accessibility contains morphisms that no element of $\mathcal{U}(\mathcal{L})$ can realize — the cross-block composition paths through hybrid sectors.

The mediating role of hybrid sectors. Every cross-block morphism in compositional accessibility must pass through a hybrid sector — a sector whose projector has nonzero restriction to multiple blocks (see Hybrid Sectors definition above). Hybrid sectors carry irreducible content from both blocks, enabling paths where Lie-generated accessibility sees only a block boundary and terminates.

7.3 The Central Formula

The T7 phenomenon is the gap between the two accessibility structures. It is not “higher-order transport” — it is **projector-mediated morphism outside Lie-generated accessibility**. The central formula of the trilogy:

$$\boxed{\mathcal{L} \subsetneq \overline{\mathcal{T}}}$$

T7 morphisms are precisely the cross-block morphisms outside Lie-generated accessibility. Concretely: S2(eo) can reach S4(ep+co) through S6 in compositional accessibility, but no element of $\mathcal{U}(\mathcal{L})$ — no Lie polynomial, no iterated commutator, no finite linear combination of products of A_g — can couple them. The structural gap decomposes cleanly by block structure: within each block, Lie-generated accessibility equals compositional accessibility. The entire gap is cross-block.

7.4 Connection to Classical Accessibility Theory

This structure reframes a classical result. The Lie algebraic rank condition (Chow 1939, Brockett 1973, Sussmann-Jurdjevic 1972) tests accessibility via iterated Lie brackets of vector fields — exactly Lie-generated accessibility. Compositional accessibility tests accessibility via finite group composition — the concatenation of generator actions through sector projectors.

For block-diagonal representations, the Lie rank condition is necessary but not sufficient for full accessibility. T7 morphisms are the morphisms that pass the compositional test but fail the Lie test — invisible to the standard accessibility criterion. **Compositional accessibility detects transport channels inaccessible to Lie-generated accessibility in the verified block-diagonal Rubik geometry.**

8 Discussion

Conditions C0–C3 and the T7 Theorem are structural: they identify the algebraic prerequisites for accessibility separation in block-diagonal finite group representations. The specific T7 data reported in this paper (5 cross-block T7 morphisms mediated through the S6–S7–S9 hub complex) is the Rubik realization. The S_3 systems serve as negative controls (C0 fails, 0 T7), not as realizations of the separation phenomenon. The present work establishes the existence of compositional accessibility beyond Lie-generated accessibility in the Rubik representation. Whether analogous separations occur in broader finite transport geometries depends on the independent verification of Conditions C0–C3.

The T7 Theorem establishes the strict separation between Lie-generated accessibility and compositional accessibility under conditions C0–C3. The proof architecture — Schur-locality (Lemma 2.3) → Lie-Generated Support Invariance (Lemma 4.1) → T7 — is structural, not numerical. The five T7 morphisms are all cross-block, all mediated through the S6–S7–S9 hub complex, and all invisible to any element of $\mathcal{U}(\mathcal{L})$. A 500-fold sweep of the transport threshold confirms the 5 T7 pairs are stable (CCS Appendix A.1), supporting the claim that T7 is not a threshold artifact.

The S_3 negative controls (nat⊕reg, reg⊕reg) provide negative controls: C0 fails in both canonical decompositions, and neither exhibits T7 — confirming C0–C3 as non-trivial characterizing conditions. What is NOT claimed and all open problems are recorded in (CCS Appendix I).

The one structural question left open is whether accessibility separation persists beyond block-preserving transport geometries.

8.1 Higher-Order Rubik Families

The verified $3 \times 3 \times 3$ Rubik cube provides the canonical realization of the $C0-C3 \Rightarrow T7$ mechanism:

$$C0 \wedge C1 \wedge C2 \wedge C3 \Rightarrow T7.$$

Preliminary structural evidence suggests that higher-order Rubik systems may exhibit analogous transport-geometry hierarchies. Coarse face-turn generators appear to create communicable sector structures and incomplete transport centers $Z \subsetneq \text{Comm}(\rho)$, while additional slice and symmetry operations subsequently activate controlled hybrid transport and cross-sector compositional accessibility.

The generator families admit a conjectural three-layer stratification:

$$\text{coarse generators} \rightarrow \text{transport activators} \rightarrow \text{compositional generators}.$$

This suggests a possible inheritance pattern:

$$C0 \rightarrow \text{transport topology} \rightarrow \mathcal{L} \subsetneq \overline{\mathcal{T}}.$$

Whether higher-order cube systems systematically exhibit T7 morphisms remains an open computational and representation-theoretic problem.

8.2 Multiplicity Interpretation of T7

The observed T7 mechanism suggests that compositional accessibility may fundamentally originate from isotypic multiplicity geometry.

Conditions C1 and C2 are **multiplicity accessibility conditions**: C1 requires that at least one irreducible representation appear across multiple sectors, and C2 requires that such multiplicity be transport-active through a hybrid sector spanning distinct blocks. Together, C1 and C2 demand that the representation possess distinguishable multiplicity — isotypic components whose internal commutant structure (noncommutative matrix blocks such as $M_2(\mathbb{C})$) enables cross-block morphisms invisible to the Lie algebra.

Under this interpretation, T7 is **distinguishable multiplicity-mediated accessibility**: compositional paths that exploit the internal multiplicity geometry of hybrid sectors to cross block boundaries that Lie-generated accessibility cannot penetrate. Hybrid transport, refinement obstruction, and T7 accessibility become different manifestations of the same underlying multiplicity structure. A general proof remains open.

8.3 Structural Defect Taxonomy of Generator Families

Exploratory. Discussion-level. Empirical pattern.

The canonical 18-generator system closes all three structural levels: arithmetic (rational spectrum), sector (9 primitive sectors), and transport (10 edges, 5 T7). Removing specific generators activates distinct failure modes. Four families were constructed by selective deletion:

n	Removed	Arithmetic Sector		Transport	Comm(ρ)
18	—	closed (Q)	closed (9)	closed (10 edges, 5 T7)	610
16	2 axis-0 HT ($\mathbb{R}^2$, $\mathbb{L}^2$)	$\sqrt{5}$ at layer	shielded (13, 2 non-rat.)	mild (32 edges, 11 T7)	610
15	3 negative-face HT	higher field	amplified (25, 4→20)	amplified (65 edges, 24 T7)	610
14	4 axis-1 QT	$\sqrt{5}$ at layer	affected (10, $\sqrt{5}$ in 2)	stable (20 edges, 5 T7)	675

$n = 16$ (**Sector Shielding**): $\mathbb{Q}(\sqrt{5})$ appears at the layer level but is largely shielded at the sector level — only 2 of 13 sectors carry non-rational λ . $n = 14$ (**Field Defect Localization**): $\sqrt{5}$ survives to the sector level but is confined to 2 EP+EO sectors; Comm(ρ) expands to 675. $n = 15$ (**Transport Resolution Amplifier**): the field becomes higher, sectors explode 9→25 with mirror splitting ($S3 \cong S6$) and hierarchical composition ($S7 \cong$ union of $S3/S6/S9$ patterns), T7 amplifies 5→24.

Across the families tested, all observed arithmetic defects and transport amplifications occur in sectors carrying EP-block support, consistent with the M_2 Principle of [1]. CP-block sectors ($S1, S8$) remain intact across all families. The comparison also suggests that transport amplification and commutant expansion are independent structural responses: $n = 15$ amplifies T7 (5→24) with unchanged Comm(ρ) = 610, while $n = 14$ expands Comm(ρ) (610→675) without T7 amplification.

This is a Discussion-level observation, not a theorem: it describes empirical regularities across four generator families. Whether the taxonomy exhausts the possible structural failure modes remains open. The complete numerical data and sector-level statistics are archived in (CCS Appendix L3).

9 Concluding Perspective

Classical accessibility theory tests reachability through the tangent structure (Chow 1939, Brockett 1973). For the Rubik representation studied here — and for finite group representations exhibiting the same block-diagonal transport geometry — the tangent picture is systematically incomplete. The universal enveloping algebra $\mathcal{U}(\mathcal{L})$ captures the infinitesimal geometry of the group orbit, but a finite group action also possesses **projector geometry**: the way sector projectors $\{P_\alpha\}$ partition the representation into subspaces that the group elements map between. This projector geometry is invisible to the tangent structure — the logarithm $A_g = \log \rho(g)$ erases it, and T7 morphisms are frozen out of $\mathcal{U}(\mathcal{L})$ entirely.

The two coincide within each invariant block and diverge at the block boundary, where only finite composition through hybrid sectors can access morphisms that the continuous limit structurally excludes:

$$\boxed{\mathcal{L} \subsetneq \overline{\mathcal{T}}}$$

is the structural signature of this incompleteness — the central result of this trilogy.

Additional spectral-triple interpretations and Dirac-operator robustness experiments are recorded in (CCS Appendix H).

9.1 Trilogy Arc

Paper	Structural Result
I	Projector geometry exists
II	Noncommutative curvature exists within projector geometry
III	Compositional accessibility exceeds Lie accessibility

Thus the continuous Lie tangent picture is structurally incomplete: finite compositional geometry contains accessibility channels invisible to the tangent algebra. Equivalently: projector geometry $\not\subseteq$ Lie tangent geometry.

Appendix A Compact Data Reference

All numerical values cited in this paper are frozen in the Unified Computational Supplement (CCS). The CCS is the authoritative numerical source; this paper reports the structural conclusions.

CCS Location	Content
CCS-I §2.3–§2.6	κ_0, κ_1 matrices; T7 morphisms; accessibility classes at 6-layer and 9-sector resolution
CCS-III	2.3 (Schur-locality), 4.1 (Lie-Generated Support Invariance), T7 Theorem (three-layer proof chain), C0–C3 analysis
CCS App. F	Empirical validation: κ hierarchy as search diagnostic framework
CCS App. G	S_3 negative controls (nat $\oplus$ reg: 9-dim, 0 T7; reg $\oplus$ reg: 12-dim, 0 T7 — negative results)

Terminology: `docs/conventions.md`. Canonical layer keys, sector data: (CCS Part 0).

Appendix B S_3 Prototypes (Negative Results)

Canonical policy. Unless otherwise stated, all sector decompositions in this trilogy are defined with respect to the transport-generated commutative algebra $Z = \langle A_{\text{full}}, A_{\text{trans}} \rangle$. Sectorizations involving auxiliary block projectors (e.g. P_{nat}) are treated as externally refined decompositions and are not considered canonical.

Sector refinement. A *refined decomposition* is obtained by adjoining auxiliary commuting projectors to the canonical transport-generated center:

$$Z_{\text{refined}} = \langle Z_{\text{canonical}}, P_{\text{aux}} \rangle$$

where $Z_{\text{canonical}} = \langle A_{\text{full}}, A_{\text{trans}} \rangle$ and P_{aux} is any projector that commutes with both A_{full} and A_{trans} (e.g., a block projector $P_{\text{nat}} = \text{diag}(I_{\dim V_A}, 0_{\dim V_B})$). Refined decompositions produce finer sectorizations (more sectors, smaller dimensions) by splitting degenerate eigenspaces of the canonical center. However, they are not unique — different choices of P_{aux} yield different sector counts — and are classified as externally refined, not canonical.

Persistence under refinement. The compositional-accessibility gap persists under externally refined sectorizations: if a cross-block pair has $K = \kappa_d = 0$ in the canonical decomposition, it remains inaccessible to Lie-generated accessibility in any refinement. Refinement can change sector boundaries but cannot create Lie-generated cross-block channels where none exist. The

structural separation between Lie-generated and compositional accessibility is invariant under the choice of commuting auxiliary projectors.

The two S_3 negative controls were investigated as test cases for the C0–C3 conditions. Under the corrected canonical decomposition (joint diagonalization eigenvector bug fixed 2026-05-26), neither exhibits T7. Full verification in (CCS Appendix G).

Appendix C Inert Hybrid Counterexample

The T7 Theorem requires C2: the hybrid sector must be **transport-active** — it must share isotypic support with sectors in both blocks. This appendix presents a minimal counterexample where a hybrid sector forms (satisfying the block-spanning criterion) but fails the transport-active criterion, confirming that hybrid block support alone is insufficient for T7 morphisms.

Consider the S_3 representation pair standard (ρ_{std} , 2-dim) vs. trivial $\oplus$ sign ($\rho_{\text{triv}} \oplus \rho_{\text{sign}}$, 2-dim). With $Z = \langle A_{\text{full}} \rangle$, eigenvalue coincidence in the commutative center produces a joint eigenspace whose projector spans both blocks — a **spectral hybrid** satisfying (i) of C2.

However, this hybrid shares no irreducible representation with sectors in both blocks simultaneously. Computing the isotypic support:

$$\text{Supp}(E_{\text{hybrid}}) = \{\text{std}\} \cap \{\text{triv}, \text{sign}\} = \emptyset$$

No single irreducible τ appears in the isotypic supports of E_α , E_γ , and E_β . The C2 criterion fails: $\text{Tr}(P_\alpha \Pi_\tau) = 0$ for all τ common to α and β — no finite composition path exists through this hybrid.

Property	S_3 nat $\oplus$ reg (0 T7)	S_3 std vs. triv $\oplus$ sign (Inert hybrid)
Hybrid exists?	Yes (S5, nat+reg)	Yes (eigenvalue coincidence)
Shared irrep?	Yes (std in both blocks)	No (std vs. triv/sign disjoint)
Transport-active?	Yes	No (inert)
T7 morphisms	0	0

This counterexample proves that C1 (shared noncommutative support) is a necessary condition: eigenvalue coincidence can produce a hybrid projector spanning both blocks without shared irrep geometry, but such an inert hybrid provides no composition path. The distinction between spectral hybrids and transport-active hybrids (see definition above) is not a definitional nuance — it is the operational boundary between T7 existence and T7 absence.

Appendix D Lie Accessibility Diagnostics

Compact summary of the κ hierarchy at 9-sector resolution. Full κ matrices in (CCS-I §2.3–§2.4); exhaustive verification in (CCS Appendix F).

D.1 Accessibility Hierarchy Summary

Depth	Object	Definition	Cross-block?	New channels
0	K (direct)	$\max_g \ P_\alpha \rho(g) P_\beta\ _F$	No	10 edges
1	κ_0 (Lie gradient)	$\max_g \ P_\alpha A_g P_\beta\ _F$	No	Preserves all $K>0$ channels
2	κ_1 (Lie curvature)	$\max_{g,h} \ P_\alpha [A_g, A_h] P_\beta\ _F$	No	7 pure curvature channels
≥ 3	κ_d (higher)	d -fold commutators	No	Amplifies within-block; no new cross-block

D.2 Pure Curvature Channels ($\kappa_0 \approx 0$, $\kappa_1 > 0$)

All 7 are within-block (share ≥ 1 block). Zero cross-block curvature channels.

Pair	κ_1	Shared block	Mechanism
S2 $\leftrightarrow$ S3	0.71	eo	Krawtchouk-order mismatch (EO)
S2 $\leftrightarrow$ S7	1.01	eo	Intra- $V_5/9$ EO coupling
S3 $\leftrightarrow$ S4	4.27	ep	Krawtchouk-order mismatch (EP, k=2 vs k=3)
S3 $\leftrightarrow$ S5	1.01	eo	Krawtchouk-order mismatch (EO)
S4 $\leftrightarrow$ S7	6.17	ep, co	Mixed EP+CO curvature
S5 $\leftrightarrow$ S7	1.42	eo	Intra- $V_5/9$ EO coupling
S7 $\leftrightarrow$ S8	6.98	cp	CP Krawtchouk-order mismatch (largest)

Enhancement ratio $\kappa_1/\kappa_0 \sim 10^{14}$ for the pure curvature channels — the signature of commutator-mediated transport where individual Lie generators are blocked.

D.3 Within-Block Restriction

Accessibility type	Within-block	Cross-block
Direct (K)	10 edges	0
Lie gradient (κ_0)	All $K > 0$ channels preserved	0
Lie curvature (κ_1)	7 pure curvature channels	0
Higher ($\kappa_d, d \geq 2$)	Amplifies existing channels	0
Compositional accessibility (T7)	0 (none needed)	5 morphisms

The within-block restriction is a structural theorem (Lemma 4.1): every Lie monomial is block-diagonal, so cross-block $\kappa_d = 0$ for all d . Curvature enriches within-block accessibility but cannot breach the block boundary.

D.4 Tolerance and Numerical Stability

Parameter	Value	Purpose
TOL	10^{-10}	General numerical assertions (projector idempotence, orthogonality)
TOL_K	0.05	Transport detection threshold ($K > \text{TOL_K} \rightarrow \text{edge exists}$)
TOL_KAPPA	10^{-6}	κ sanity consistency floor (logm noise ceiling $\sim 10^{-8}$)
logm fidelity	$< 3 \times 10^{-8}$	$\max_g \ \exp(A_g) - \rho(g)\ $ — principal matrix logarithm accuracy

logm numerical leakage. `scipy.linalg.logm` produces off-diagonal errors $\sim 10^{-6}$ to 10^{-8} across adjacent blocks (e.g., $\text{co} \leftrightarrow \text{eo}$ at indices 208–228). These are numerical artifacts, not structural κ signals. The structural T7 test (block-set disjointness $\implies \kappa_d = 0$) is immune to this leakage; $\text{TOL_KAPPA} = 10^{-6}$ serves as a consistency sanity check, not the primary detection criterion.

Bibliographic Notes

Mathematical lineage. This paper belongs to the tradition of **Lie algebraic accessibility and geometric control theory** — the Chow-Brockett-Jurdjevic-Sussmann program of understanding nonlinear accessibility through Lie brackets of the accessibility algebra. The κ_d

hierarchy (§3) is a discrete-group analogue of the iterated Lie-bracket accessibility distributions; the T7 Theorem (§5) establishes that, on the block-decomposed representations studied here, discrete composition realizes morphisms that no element of the universal enveloping algebra $\mathcal{U}(\mathcal{L})$ produces — a structural mismatch that does not arise in the continuous theory. The lineage runs: nonholonomic systems and bracket-generating distributions (Chow 1939) $\rightarrow$ Lie algebraic accessibility and orbit theorems (Brockett 1973, Sussmann-Jurdjevic 1972) $\rightarrow$ geometric control on Lie groups (Jurdjevic 1997) $\rightarrow$ finite-dimensional representations and universal enveloping algebras (Serre 1977, Humphreys 1972) $\rightarrow$ the present paper.

Lie algebraic accessibility and geometric control

[1] W.-L. Chow, “Über Systeme von linearen partiellen Differentialgleichungen erster Ordnung.” *Mathematische Annalen* 117:98–105, 1939. — The foundational result that a bracket-generating distribution makes the orbit equal to the ambient manifold (Chow-Rashevskii theorem). The block-preserving Lemma 4.1 is the opposite phenomenon for our representation: the Lie algebra $\mathcal{L} = \text{Lie}\{A_g\}$ is *not* bracket-generating across blocks, so the Lie orbit is strictly smaller than the compositional reach.

[2] R.W. Brockett, “Lie Theory and Control Systems Defined on Spheres.” *SIAM Journal on Applied Mathematics* 25(2):213–225, 1973. — Foundational paper for Lie-algebraic controllability: Lie brackets of drift and control vector fields generate the accessibility algebra. The κ_d hierarchy is the discrete-group analogue — κ_0 from individual A_g , κ_1 from first brackets $[A_g, A_h]$, κ_d from depth- d commutators.

[3] H.J. Sussmann and V. Jurdjevic, “Controllability of Nonlinear Systems.” *Journal of Differential Equations* 12:95–116, 1972. — The orbit theorem: the reachable set from a point is the immersed submanifold whose tangent space is the accessibility algebra evaluated at that point. The central T7 question — “what can the discrete composition reach beyond the Lie orbit?” — is the discrete counterpart.

[4] V. Jurdjevic, *Geometric Control Theory*. Cambridge University Press, 1997. — Comprehensive treatment of Lie-algebraic controllability on Lie groups: drift fields, control fields, accessibility distributions, and the Chow-Rashevskii theorem. The continuous-control framework against which T7 is measured and shown to be strictly richer.

[5] H. Nijmeijer and A.J. van der Schaft, *Nonlinear Dynamical Control Systems*. Springer, 1990. — Accessibility distributions and the Lie algebraic rank condition for nonlinear control systems. The iterative construction of the accessibility algebra by Lie-bracket closure is the continuous ancestor of the κ_d depth hierarchy.

Universal enveloping algebra and finite-group representations

[6] J.-P. Serre, *Linear Representations of Finite Groups*. Graduate Texts in Mathematics 42, Springer, 1977. — Schur’s Lemma, isotypic decomposition, and the structure of $\mathbb{C}[G]$ -modules. The block decomposition $V = V_A \oplus V_B$ and the block-preserving property of $\mathcal{L}$ (Lemma 4.1) are consequences of the isotypic structure of ρ . The T7 condition C1 — shared isotypic support — is the necessary intertwiner obstruction read off from this decomposition.

[7] J.E. Humphreys, *Introduction to Lie Algebras and Representation Theory*. Graduate Texts in Mathematics 9, Springer, 1972. — Standard reference for universal enveloping algebras $\mathcal{U}(\mathcal{L})$, the Poincaré-Birkhoff-Witt theorem, and the relation between Lie-algebra and group-algebra representations. The statement “ $\overline{\mathcal{T}} \setminus \mathcal{L}$ contains morphisms unreachable by any element of $\mathcal{U}(\mathcal{L})$ ” (see the Structure Behind T7) is the categorical reformulation of Lemma 4.1 in this language.

Category-theoretic language

[8] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed. Graduate Texts in Mathematics 5, Springer, 1998. — Standard reference for the elementary categorical notions used in the Structure Behind T7 section (category, subcategory, faithful functor, compositional closure). The transport category $\overline{\mathcal{T}}$ and the Lie sublayer $\mathcal{L}$ are used as structural language to compare two generating systems; no advanced category theory is invoked.

Trilogy cross-references

[2], [1], and [3] are defined in `papers/tex/trilogy.bib`.

Code Availability

All numerical experiments, projector constructions, transport computations, and figure-generation scripts are available at:

<https://github.com/dooven-prime/rime-lite>

The repository also contains the Unified Computational Supplement (CCS), canonical datasets, and reproducibility notebooks corresponding to the trilogy.

References

- [1] WuJun Chen. Noncommutative transport topology in the Rubik’s cube representation: Hybrid sectors, permutation channels, and refinement obstructions. *arXiv preprint*, 2026. Paper II of the RIME trilogy.
- [2] WuJun Chen. Spectral sector decomposition in the Rubik’s cube representation: Rational spectral collapse, primitive idempotents, and block spectral factorization. *arXiv preprint*, 2026. Paper I of the RIME trilogy.
- [3] WuJun Chen. Unified computational supplement for Papers I–III, 2026. Definitive numerical constitution for the RIME trilogy. Available at <https://github.com/dooven-prime/rime-lite>.